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Reviews in Medical Virology

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Forecasting Influenza Epidemics and Pandemics in the Age of AI and Machine Learning

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Received: 11 November 2025 | **Revised:** 31 December 2025 | **Accepted:** 14 January 2026

ABSTRACT

Influenza's rapid evolution, driven by its segmented RNA genome, high mutation rate, and extensive animal reservoirs, underpins its capacity to cause recurring epidemics and unpredictable pandemics. Recent advances in artificial intelligence (AI) and machine learning (ML) are transforming influenza forecasting by enabling the prediction of viral evolution and the optimisation of public health preparedness. This review synthesises insights from historical data (1890–2025) and contemporary research to examine the evolving role of AI in influenza prediction. It highlights major developments including transformer-based models for viral evolution, real-time integration of mobility and environmental data, hybrid quantum, which are classical algorithms, and multimodal data fusion frameworks, it also consideres critical risk modifiers such as meteorological variation, armed conflict, and host genetics. Importantly, the review distinguishes between retrospective, proof-of-concept analyses and prospective, real-time forecasting applications, clarifying their respective contributions to operational public health preparedness and informed decision-making.

1 | Introduction

Accurate and timely forecasting is crucial for the effective management of influenza outbreaks, facilitating a shift from reactive responses to proactive public health interventions [1]. Advances in artificial intelligence (AI) and machine learning (ML) have revolutionised epidemiological modeling, enabling the prediction of epidemic trajectories, real-time monitoring of viral evolution, and the rapid deployment of targeted control measures [2]. These technologies leverage complex data streams to capture the multifaceted nature of influenza transmission, incorporating biological determinants such as viral genetics and host immunity, environmental influences including meteorological variables and ultraviolet radiation, as well as social factors like human mobility and behavioural patterns (Figure 1) [3].

This review examines the integration of AI and ML methodologies within influenza forecasting frameworks, focussing on the fusion of heterogeneous data types through advanced predictive analytics. Notably, recent developments in quantum computing and multimodal data integration have demonstrated significant potential to enhance computational efficiency and model accuracy. These approaches enable the simultaneous analysis of genomic sequences, environmental parameters, and epidemiological indicators, thereby strengthening the spatiotemporal precision of outbreak predictions.

This schematic illustrates a multifaceted machine learning framework that integrates antigenic drift prediction, antigen mapping, immunity dynamics, host factors (age, sex, genetics), meteorological data, and conflict zone analysis to

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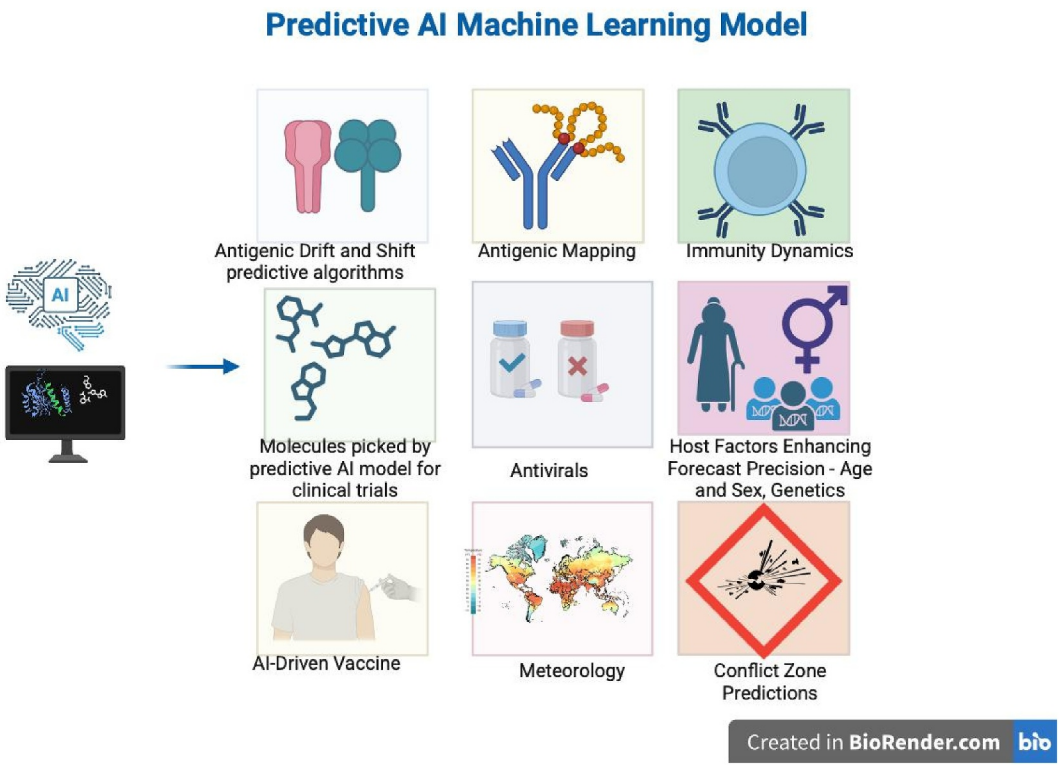


FIGURE 1 | A comprehensive predictive AI model for infectious disease response.

support the development of antiviral agents, AI-guided vaccine strategies, and the selection of candidate molecules for clinical trials.

Recent progress in computational epidemiology underscores the capacity of AI systems to integrate diverse data sources, encompassing genomic surveillance, real-time mobility, and climate metrics [4]. The amalgamation of these datasets has led to refined temporal and spatial resolution in forecasting models, enabling earlier detection of transmission hotspots and more precise estimation of epidemic peaks [5]. Furthermore, the deployment of adaptive learning algorithms facilitates continual model recalibration, allowing forecasts to dynamically incorporate new information about pathogen evolution and shifting population susceptibility [6]. Such responsiveness is crucial in the context of rapidly evolving viral landscapes and diverse public health interventions, ultimately leading to more effective disease control and mitigation strategies [7].

This review provides a distinct contribution by systematically differentiating between retrospective, proof-of-concept AI models and prospective, real-time forecasting applications relevant to public health decision-making. In contrast to prior reviews that primarily focus on algorithmic performance, we integrate biological, environmental, social, and conflict-related determinants within a unified analytical framework. By critically evaluating the operational readiness, limitations, and contextual applicability of AI-driven influenza forecasting, this work aims to bridge the gap between methodological innovation and real-world public health implementation.

1.1 | Literature Search and Selection

Relevant publications were identified through searches in PubMed, Scopus, and Web of Science covering the period from 2000 to 2025, using the keywords ‘influenza forecasting’, ‘machine learning’, ‘artificial intelligence’, ‘epidemiological modeling’, ‘viral evolution’, and ‘predictive analytics’. Additional sources were retrieved from reference lists and preprint repositories (medRxiv, arXiv) to include emerging research. Studies were included if they applied or evaluated AI or ML approaches for influenza prediction, surveillance, or vaccine design.

Emphasis was placed on studies describing model architecture, data sources, and performance metrics, with explicit notation of whether analyses were conducted retrospectively or prospectively. This strategy ensured balanced coverage of both conceptual and applied advances in AI-driven influenza forecasting.

Studies were excluded if they lacked sufficient methodological detail, relied solely on simulated data without epidemiological grounding, or did not report the outcomes of model validation. Given the heterogeneity of study designs, a formal meta-analysis was not performed; instead, a structured narrative synthesis was

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applied. This approach allowed for a balanced assessment of methodological rigor, data sources, and practical relevance across diverse AI-based forecasting studies.

2 | Historical Perspective: Lessons Shaping Modern Forecasting

Historical pandemics have informed today's AI-driven forecasting tools. The 1890 epidemic revealed early transmission patterns [8], while the 1918 H1N1 pandemic, with over 50 million deaths [9], spurred basic surveillance systems—ancestors of modern predictive models. Later outbreaks (1957 H2N2, 1968 H3N2, 2009 H1N1pdm09) refined these approaches by exposing viral spread dynamics [10]. The drastic decline in influenza cases during COVID-19's non-pharmaceutical interventions (NPIs) in 2020–21 [11] underscored the power of real-time data, now supercharged by AI.

Recent studies have leveraged historical pandemics to enhance the performance of AI models. Recent studies employing graph neural networks (GNNs) have demonstrated improved spatial influenza nowcasting, though direct re-analysis of early pandemic data remains limited [12]. The 2009 H1N1pdm09 pandemic highlighted the role of global air travel, prompting machine learning (ML) models to incorporate flight data, which reduced prediction errors [13]. The 2020–21 influenza decline during COVID-19 NPIs provided a dataset for training LSTM models, achieving an accuracy boost in low-transmission scenarios [14]. Additionally, transfer learning from historical pandemics has enabled models to generalise across diverse influenza strains, with evidence showing an improvement in predicting the dynamics of novel strains [15].

3 | Biological Drivers in Forecasting Models

Influenza's biology underpins predictive algorithms: antigenic drift involves mutations in hemagglutinin (HA) and neuraminidase (NA) at approximately $2–3 \times 10^{-3}$ substitutions per site per year, which challenge existing immunity [16]. Recent transformer-based and sequence-learning approaches have shown promise for predicting antigenic drift from genomic data, which may help anticipate vaccine mismatch [17–19]. Antigenic shift, resulting from reassortment events in animal reservoirs, leads to the emergence of novel influenza strains, with deep learning techniques identifying shift risks through mapping of co-infection zones [20]. Additionally, immunity dynamics, characterised by short-lived immune protection especially pronounced in older adults [21], are incorporated into age-specific forecasting models to improve prediction accuracy.

Biological factors play a crucial role in influenza forecasting through the analysis of virus-host protein interactions and pathogenicity prediction. For instance, the use of XGBoost to predict influenza A virus-human protein-protein interactions has demonstrated high accuracy [22]. Machine learning models based on the predicted structure of hemagglutinin enable the assessment of avian influenza pathogenicity [23]. Furthermore, models focused on antigenic distance prediction [17, 24] and zoonotic potential assessment [25] underscore the importance of biological drivers in forecasting.

These biological mechanisms represent key uncertainties in forecasting, primarily due to the non-linear nature of viral evolution across geographic and host boundaries [26]. Advances in computational virology have enabled the integration of host-pathogen interactions into dynamic models, capturing the probabilistic nature of antigenic events [27]. Predictive tools now incorporate protein structural data to detect functionally significant mutations with immunological consequences. The inclusion of zoonotic interface data from agricultural systems has expanded forecasting scope beyond human populations [28]. Furthermore, real-time genomic surveillance pipelines have shortened the response window for identifying threats related to drift or shifts [29].

Genomic sequencing advancements have revolutionised biological forecasting. Recent ML models integrating sequence-based features have achieved high accuracy in predicting antigenic distances, offering insights into antigenic drift [17, 18]. Another study in 2024 introduced a multi-modal ML approach integrating metagenomic data from environmental samples (e.g., poultry farms), which improved shift predictions [30]. Immunity dynamics models now incorporate longitudinal serology data, with a 2024 study using Bayesian machine learning (ML) to predict waning immunity in elderly populations, thereby boosting age-specific forecast accuracy [31].

4 | Environmental Inputs: Meteorology and Solar-UV in Predictions

Environmental determinants are discussed in this section in an integrated manner to avoid repetition across subsequent methodological and application-focused sections. Meteorological and solar ultraviolet factors are therefore addressed here as cross-cutting modifiers of influenza transmission, informing rather than duplicating later discussions of AI model architectures and performance.

They represent critical environmental components of influenza forecasting models and play a critical role in refining forecast accuracy. Low absolute humidity, approximately 4 g/m^3, enhances viral survival and transmission potential [32]. AI models incorporate real-time weather data, such as humidity and temperature, to more precisely predict transmission peaks. Solar ultraviolet (UV) radiation also significantly influences influenza dynamics; a UV-Index below 2 is associated with increased influenza surges, while each unit increase in UVI reduces transmission rates by 7%–10% [33]. Neural network models leverage these UVI trends to adjust seasonal forecasts, projecting an extension of influenza seasons by 2050 in response to anticipated climate change [34, 35].

Incorporating meteorological data and solar ultraviolet (UV) radiation into influenza forecasting models has been shown to enhance the precision of epidemiological predictions significantly. While the number of studies explicitly addressing these environmental determinants remains limited in the existing

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body of literature, recent advances in predictive modeling underscore the importance of integrating climate-related variables, such as temperature, humidity, and solar UV indices, into surveillance frameworks. This approach not only improves the temporal and spatial resolution of outbreak forecasts but also contributes to a more comprehensive understanding of the complex interplay between environmental conditions and viral transmission dynamics [12, 36, 37].

Recent advances in atmospheric data resolution have improved the spatial precision of influenza forecasting, particularly in regions with previously limited monitoring capabilities [38]. Machine learning frameworks now integrate not only meteorological indicators but also microclimatic fluctuations to capture short-term variability in transmission risk. The inclusion of dynamic environmental baselines enables more robust adaptation of models to climate anomalies and extreme weather patterns [39].

Integration of meteorological and satellite data in convolutional neural network frameworks has been shown to improve short-term influenza forecasts in tropical regions [40, 41]. Emerging evidence indicates that increased levels of ultraviolet radiation are associated with decreased influenza transmission and supports the application of machine learning models to predict UV-driven seasonal dynamics with substantial accuracy [17]. NeuralGCM models project a 10%–15% extension of influenza seasons by 2050 due to climate change [35, 42]. Air pollution data (e.g., PM2.5 levels) have also been incorporated into AI systems, resulting in an increase in transmission observed in high-pollution zones, which enhances urban-focused forecasting [43]. Additionally, ensemble machine learning models that combine meteorological, UV, and climate variables have improved long-term forecasting performance. However, data latency in remote and low-resource regions remains a limiting factor [38].

5 | AI and Machine Learning: Core of Influenza Forecasting

AI powers advanced surveillance and prediction through several key methodologies. Nowcasting models, such as LSTM and gradient-boosting algorithms, integrate syndromic, laboratory, and mobility data to reduce forecasting errors by 10%–25% over 1–8-week horizons [27, 30]. The inclusion of real-time smartphone mobility data further enhances the precision of these short-term forecasts [44]. Early warning systems employ natural language processing (NLP) techniques to analyse news reports and travel data, enabling the detection of outbreaks like the 2019 H1N1pdm resurgence prior to official confirmation [45]. Transformer models are utilised to predict viral evolution by analysing genomic sequences, facilitating anticipatory responses [19]. In antigenic mapping, deep learning approaches account for approximately 80% of the variance in H3N2 strains, thereby accelerating vaccine strain selection processes [17]. Additionally, graph neural networks contribute by forecasting cross-strain immunity patterns, which informs broader vaccine design strategies [46].

A range of machine learning architectures, including deep recurrent, graph-based, and transformer models, have been applied to influenza prediction tasks using diverse datasets. Their comparative features and reported accuracies are summarised in Table 1.

Recent developments in quantum machine learning (QML) and multi-modal data fusion have pushed the boundaries of influenza forecasting [52]. Emerging hybrid quantum-classical algorithms have been proposed to accelerate high-dimensional modeling tasks in infectious disease research, though their application to influenza forecasting remains largely conceptual [52, 53]. Enhanced natural language processing models, utilising multilingual social media datasets, have enabled the detection of outbreaks several days earlier than traditional surveillance systems. Graph-based deep learning architectures have shown potential for modeling cross-strain immune interactions [12, 28], though their validation in influenza forecasting is still preliminary [12, 18]. Multi-modal fusion models combining genomic, syndromic, and social media data have achieved significant reductions in nowcasting errors, underscoring the value of integrated approaches. These advancements underscore the increasing potential of AI to provide timely, precise, and context-aware influenza forecasts.

TABLE 1 | Representative AI and machine learning models applied in influenza forecasting.

Model/Algorithm	Primary data Type(s)	Forecast time horizon	Representative reference
LSTM (long short-term memory)	Epidemiological, mobility, meteorological	1–8 weeks ahead	[47]
Graph neural network (GNN)	Spatial-temporal case data, mobility, climate	Real-time and 1–4 weeks forecasts	[48]
Transformer-based sequence model	Genomic (HA/NA), antigenic data	Seasonal to multi-year	[18]
CNN (convolutional neural network)	Meteorological + satellite data	1–6 months	[49]
XGBoost ensemble	Epidemiological + host factors	2–4 weeks	[50]
Reinforcement learning (RL)	Epidemiological + intervention data	Scenario simulation	[51]
Multimodal fusion model	Genomic + social media + mobility	Real-time	[37]

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Collectively, these models highlight the complementary strengths of neural, statistical, and hybrid approaches in enhancing temporal and spatial forecasting precision across diverse data environments. Continued innovation in algorithmic methods and data integration will be crucial to meeting the challenges of evolving viral threats [54].

Together, the summarised models (Table 2) and the end-to-end workflow (Figure 2) outline how AI frameworks transform complex, multi-source data into actionable public health intelligence.

The diagram illustrates six interconnected stages — from data acquisition and integration to decision support — highlighting the transformation of heterogeneous datasets into actionable insights for public health and vaccine strategy optimisation.

It is essential to differentiate between retrospective model evaluations, where historical datasets are used to benchmark algorithmic accuracy, and prospective, real-time implementations designed to guide ongoing public health decisions. Most of the AI and ML models summarised in Table 1 have been developed and validated retrospectively, using previously collected genomic, epidemiological, or environmental data. These studies often demonstrate high accuracy in back-testing scenarios but may overestimate actual performance in real-world settings.

In contrast, several prospective implementations have applied AI forecasting in operational settings. Real-time nowcasting systems that integrate mobility and syndromic data, as well as NLP-based outbreak detection tools, have supported situational awareness in CDC and WHO surveillance frameworks [59, 60].

TABLE 2 | Representative data types used in influenza forecasting and related preprocessing challenges.

Data type	Examples of sources	Forecasting contribution	Key preprocessing challenges	Representative references
Genomic data	GISAID, NCBI influenza virus resource	Tracking antigenic drift/shift, predicting vaccine mismatch	Sequence alignment, removal of low-coverage reads, metadata harmonisation	[55]
Epidemiological data	WHO FluNet, CDC ILINet, local surveillance reports	Case forecasting, early outbreak detection	Reporting delays, underreporting, inconsistent time resolution	[56]
Meteorological and UV data	NOAA, Copernicus climate data store	Modeling seasonality, linking humidity and temperature to transmission	Spatial interpolation, normalisation of variables, missing data handling	[40]
Mobility and population data	Google mobility reports, airline network data, Facebook data for good	Modeling transmission pathways, import/export risks	Privacy-preserving aggregation, regional normalisation	[57]
Satellite imagery and environmental data	NASA MODIS, Sentinel-2, VIIRS	Identifying climate and land-use correlates of outbreaks	Cloud masking, georeferencing, temporal resampling	[41]
Social media and NLP-derived data	Twitter, Weibo, news APIs	Early detection via syndromic and sentiment signals	Language diversity, noise filtering, bot detection	[58]

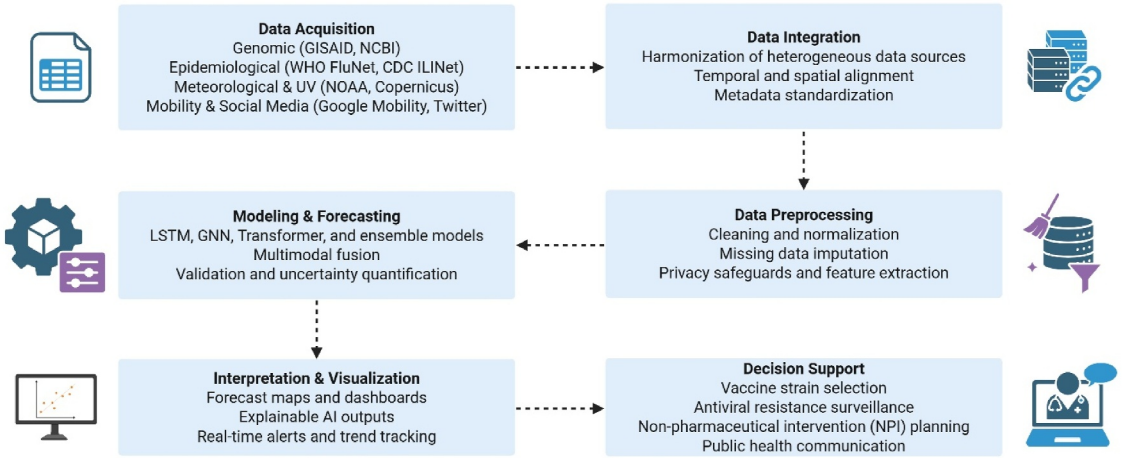


FIGURE 2 | Workflow of AI-driven influenza forecasting and decision support.

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However, these systems typically display higher uncertainty due to data latency, underreporting, and evolving epidemic conditions [61, 62]. Clarifying whether models were used retrospectively or prospectively is crucial for assessing the actual public health value and readiness of AI-driven forecasting technologies [62]. This distinction also highlights the importance of transparency in model validation and evaluation standards for real-world applications.

5.1 | Common Datasets and Data Preprocessing Challenges in Influenza Forecasting

AI-based influenza forecasting relies on the integration of heterogeneous datasets that capture viral, environmental, and behavioural dynamics [63]. Table 2 summarises the major data sources used in recent studies and highlights typical preprocessing challenges. Harmonising these diverse inputs remains a crucial step for improving model reliability and reproducibility.

Standardisation of metadata, development of federated learning frameworks, and automated feature extraction pipelines are emerging as key strategies to overcome these challenges and facilitate large-scale, real-time forecasting.

Reproducibility remains a major challenge in AI-based influenza forecasting, as many studies rely on proprietary datasets or lack publicly available code and standardised evaluation benchmarks. Greater adoption of open data practices, shared preprocessing pipelines, and federated learning frameworks would substantially improve transparency and facilitate independent validation across diverse settings.

5.2 | Limitations and Challenges of Machine Learning in Influenza Forecasting

Despite significant progress, several limitations constrain the practical implementation of AI and ML in influenza forecasting.

First, overfitting remains a major issue, particularly when models are trained on limited or region-specific datasets. Highly complex architectures such as deep neural networks can achieve excellent retrospective performance but may fail to generalise to unseen outbreaks or geographic regions [64, 65].

Second, interpretability challenges hinder the adoption of black-box models in public health decision-making. Many forecasting systems provide accurate outputs without transparent reasoning, making it difficult for epidemiologists to assess the reliability of predictions or to justify interventions based on them [66].

Third, regional data scarcity and heterogeneity limit global model transferability. Forecasting systems trained on data from high-income countries often perform poorly in low- and middle-income regions due to inconsistent reporting, lack of genomic sequencing, and differences in health infrastructure [67].

Additionally, data latency, privacy constraints, and biases from social media or mobility datasets can distort early warning outputs. Addressing these issues requires hybrid frameworks that combine mechanistic and data-driven models, robust validation pipelines, and explainable AI methods to ensure interpretability and reproducibility across diverse settings [68].

A growing body of evidence suggests that the most reliable influenza forecasts are derived from the integration of AI-driven, statistical, and mechanistic models, rather than relying on a single methodological paradigm. Ensemble frameworks that combine outputs from machine learning algorithms with compartmental (e.g., SEIR-type) or autoregressive models enhance forecast accuracy, stability, and uncertainty quantification, especially during atypical transmission periods. These hybrid approaches unite the interpretability and theoretical grounding of mechanistic models with the adaptive learning capacity of AI, yielding complementary strengths for real-time prediction and policy planning.

However, the published literature often overrepresents successful or high-performing AI models, introducing a potential publication bias that obscures null or less favourable results. Notably, several studies have shown that machine learning approaches do not consistently outperform traditional models when applied prospectively, particularly in data-sparse settings or during irregular influenza seasons [69, 70]. In such contexts, simpler autoregressive or compartmental frameworks may achieve comparable, or even superior, real-time accuracy. Acknowledging these limitations is essential to maintain a balanced perspective on AI's role and to ensure that future research builds on realistic expectations of predictive performance.

Beyond technical limitations, ethical and governance challenges remain central to the deployment of AI-driven influenza forecasting. The use of mobility data, social media signals, and satellite imagery raises concerns regarding privacy, data ownership, and potential misuse of surveillance. Moreover, unequal access to high-quality data and computational resources may exacerbate existing global health disparities, underscoring the need for transparent governance frameworks and ethically grounded deployment strategies.

6 | Translating Forecasts Into Vaccine and Antiviral Strategies

Influenza forecasting not only anticipates epidemic trends but also directly informs the design and update of vaccines and virus-directed antivirals [71, 72]. Predictive models that track antigenic drift and reassortment guide the timely selection of vaccine strains, improving alignment with circulating viruses [73]. Machine learning algorithms integrating genomic and epidemiological data enable real-time evaluation of vaccine effectiveness and prediction of resistance patterns [74].

By linking genomic forecasts with immunological and epidemiological datasets, AI systems can identify emerging mutations that reduce vaccine efficacy, providing early warnings for

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formulation updates [75, 76]. These approaches also support the anticipation of antiviral resistance, optimising therapeutic strategies before widespread clinical failure occurs [77].

Recent advances in transformer-based and multimodal models have strengthened this connection between forecasting and response, allowing rapid adaptation of vaccine composition and deployment strategies [78, 79]. In this sense, AI-driven innovation serves as a practical extension of forecasting—transforming predictive insights into actionable public health interventions that enhance preparedness and resilience against evolving influenza threats.

7 | Forecasting Amid Conflict

War disrupts traditional disease surveillance systems, yet artificial intelligence demonstrates adaptability in these challenging environments. By leveraging satellite imagery, mobility data, and natural language processing (NLP) of open-source information, AI tools effectively monitor influenza transmission among displaced populations. This integration enables the timely forecasting of outbreaks, providing critical guidance for humanitarian aid and public health interventions in conflict zones [44, 80].

Forecasting influenza outbreaks in conflict zones poses challenges due to data instability and dynamic conditions. Although direct studies on this issue are scarce in the current literature, the use of flexible and multi-source data models, such as graph neural networks capturing temporal, geographical, and functional spatial features, may enhance forecast robustness in such complex environments [12].

These challenges are exemplified by the ongoing Russia-Ukraine war, where the degradation of healthcare infrastructure has further complicated disease monitoring and response efforts. The Russia-Ukraine war, escalating in February 2022, has severely disrupted formal influenza surveillance [81]. This led to a near-complete cessation of formal infectious disease reporting, including influenza, in conflict-affected regions [81]. AI-driven open-source intelligence platforms have mitigated this by analysing multilingual news and social media [81]. These platforms have enabled the early detection of outbreaks in displaced populations, with reports indicating a decrease in traditional influenza reporting, necessitating a reliance on alternative data sources [82]. AI models that integrate satellite imagery and mobility data have improved outbreak predictions in war-torn areas, although challenges remain due to restricted access and data sparsity [83].

One such region where these challenges became particularly evident is southern Ukraine, following the collapse of the Kakhovka Dam. The collapse of the Kakhovka Dam released approximately 18 km^3 of water, flooding over 620 km^2 and affecting more than 100,000 people. This event mobilised an estimated 83,000 tonnes of heavy metals from reservoir sediments, contaminating water sources. Disrupted sewage and flooding increased waterborne disease risks, while dislodged landmines restricted access to care and mobility data. AI models integrating satellite and water data improved outbreak forecasting. Damaged irrigation systems have also exacerbated food insecurity and influenza vulnerability [84–88].

In response to these compounding vulnerabilities—including rising rates of chronic illness, comorbidities, depression, and disrupted gut microbiota—recent AI-driven innovations are increasingly being deployed to support disease surveillance and prediction in high-risk, resource-limited settings [89, 90]. AI advancements are bridging gaps in conflict zones. The integration of drone-based mobility data with graph neural networks has improved predictions of influenza spread in refugee camps. Natural language processing models that incorporate local dialects have enhanced early warning capabilities. Satellite imagery, combined with machine learning techniques, has been utilised to estimate population densities in conflict-affected areas, leading to enhanced outbreak forecasting. Furthermore, emerging blockchain-based health information systems may improve secure data sharing in resource-limited or unstable settings, though evidence of practical deployment during conflicts remains limited. These innovations demonstrate the potential for AI to overcome traditional surveillance limitations in conflict settings, providing critical support for public health interventions under challenging conditions [91–94].

8 | Host Factors Enhancing Forecast Precision

Demographic and genetic factors significantly enhance the accuracy of influenza forecasting models. Age and sex are critical determinants, as children and elderly populations often drive both transmission and disease severity [95]. Sex-based immune differences, including hormonal influences such as estrogen-mediated effects, further refine these models by capturing variations in immune response between males and females. Additionally, genetic variants like IFITM3 rs12252-C have been linked to a higher risk of severe influenza outcomes [96]. The integration of multi-omics data—comprising single-nucleotide polymorphisms (SNPs) and transcriptomic profiles—through machine learning methods improves predictions of individual susceptibility and disease progression, thereby contributing to more precise and personalised forecasting [97].

Host-related factors, including laboratory parameters, significantly improve diagnostic and predictive accuracy. Machine learning models utilising laboratory data have shown efficacy in predicting influenza A and B infections [98] and forecasting hospitalisations [99], highlighting the value of host-specific information in surveillance.

Multi-omics has advanced host factor integration. Recent research combining genomic, proteomic, and metabolomic data has improved susceptibility predictions [100]. Sex-specific models that account for hormonal influences have enhanced severity predictions in females [101]. Additionally, data from wearable devices, such as heart rate variability, have increased the accuracy of early infection detection [102]. Incorporating lifestyle factors and comorbidities into machine learning models shows promise for further refining risk assessments. The integration of host factors with environmental and viral data is

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emerging as a key strategy to enhance model robustness. Efforts to standardise data collection and ensure privacy are critical for the broader application of these tools.

9 | Public Health Decision Support

AI scenario engines turn forecasts into action by transforming predictive insights into targeted public health responses. NPI tuning leverages real-time environmental and mobility data to dynamically optimise local interventions such as school closures, travel restrictions, or mask mandates, enhancing their effectiveness while minimising disruption [103]. Vaccine strategy models integrate transmission forecasts with demographic and risk factor data to prioritise high-risk regions, enabling more equitable and efficient distribution of limited vaccine supplies, especially during early outbreak phases [104].

While retrospective analyses demonstrate the impressive theoretical capacity of AI to capture influenza dynamics, operational, real-time applications remain comparatively rare and methodologically constrained. Initiatives such as the CDC's FluSight Network and WHO's digital epidemic intelligence platforms have begun to incorporate machine-learning-based forecasts to inform vaccine allocation and non-pharmaceutical interventions. However, their predictive accuracy in live environments often declines relative to retrospective testing due to incomplete or delayed surveillance data, differences in national reporting systems, and the stochastic nature of ongoing epidemics.

These real-world limitations highlight the importance of adaptive model recalibration, uncertainty quantification, and integration with mechanistic epidemiological models. Continuous validation under operational conditions is therefore critical to ensure that AI-derived forecasts provide tangible, reliable benefits for decision-makers during active outbreaks.

Accurate influenza forecasts and early outbreak detection underpin effective public health decision-making. Deep learning models analysing search engine data and other multisource inputs enable real-time epidemic trend monitoring [105, 106]. Ensemble machine learning methods provide precise hospitalisation predictions [99], while comparative studies demonstrate the utility of machine learning algorithms in assessing hospital-acquired influenza risk [50].

A reinforcement learning framework for optimising NPIs demonstrated a reduction in case numbers in simulated environments [107, 108]. Vaccine allocation models that integrate socioeconomic indicators have improved equitable distribution in resource-limited settings [109, 110]. Digital twin models enable real-time NPI simulations, facilitating dynamic policy adjustments to health outcomes [111]. Federated learning approaches have promoted global data sharing and collaboration, enhancing model robustness [112].

10 | Conclusion

Artificial intelligence and machine learning can enhance influenza preparedness by enabling earlier detection and more adaptive forecasting. However, their value in practice remains limited by inconsistent data quality, model transparency, and algorithmic bias. Current evidence shows that AI methods do not consistently outperform established statistical or mechanistic models, highlighting the need for rigorous and balanced evaluation.

Future research should prioritise the prospective validation of real-time, interpretable systems and explore their integration with ensemble and mechanistic frameworks to strengthen reliability and policy relevance. Transparent and ethically governed data infrastructures are crucial for ensuring equitable access, especially in settings with limited digital capacity.

While AI and machine learning offer substantial potential to enhance influenza forecasting, current evidence suggests that their real-world performance remains highly context-dependent. In several prospective applications, AI-based approaches do not consistently outperform traditional statistical or mechanistic models, particularly in data-scarce or rapidly evolving settings. Recognising these limitations is essential to avoid overreliance on algorithmic outputs and to promote responsible, evidence-based integration into public health practice.

Developing standardised benchmarks, open data sharing, and cross-sector collaboration will help move the field beyond proof-of-concept studies. These efforts are crucial to ensuring that predictive technologies evolve into trusted, evidence-based tools that support global public health decision-making.

Future progress in AI-driven influenza forecasting will depend on prospective validation, standardised benchmarking, and the development of ethically governed data infrastructures. Strengthening reproducibility, interpretability, and cross-sector collaboration will be critical to transforming AI models from experimental tools into trusted components of global influenza preparedness and response.

Author Contributions

Conceptualisation: D.K., O.K., software and data curation: D.K., O.K, writing – original draft preparation: D.K., I.H., I.K., V.O., writing – review and editing, all authors. visualisation, P.P., O.K., D.K. supervision, O.K., D.K. project administration, D.K., O.K. All authors have read and agreed to the published version of the manuscript.

Ethics Statement

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The authors have nothing to report.

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